GMI German-Malaysian Institute

DIPLOMA PROGRAMME

Practical Test

Academic Period: July 2024

Code	SPG 0542	Course Name	BASIC OF PROGRAMMING
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No	Assessment Criteria	CLO	PO	Level of Diff	Full Marks	Score	Marks	Comment
1.	Insert marks based on no of student entered				5			
2.	Display menu [4 options]				5			
3.	Display highest mark				10			
4.	Display lowest mark				10			
5.	Display average marks				10			
6.	Apply array statement				5			
7.	Apply function statement				5			
				Total	**50**			

	Total Marks	
Total	**PO**	
	PO	
	Signature	

Answer all questions.

- Write a C program to allow users to input marks for multiple students and store them in an array.
- Implement a menu interface at least four options, including an exit option.
- Provide functionality to find the lowest marks among the entered marks.
- Provide functionality to find the highest marks among the entered marks.
- Provide functionality to find the average marks among the entered marks.
- Utilize function statements to implement the functionality for finding the lowest, highest, and average marks.
- Test the program to ensure correct functionality and efficiency.
- You can refer sample output given below:

```
Hello all, Welcome to Lab Task Session!

########################################
Enter the number of students: 3
Enter the marks for each student:
Enter marks for student 1: 56
Enter marks for student 2: 67
Enter marks for student 3: 90

Menu:
1. Find lowest mark
2. Find highest mark
3. Find average mark
4. Exit
Enter your choice: 1
The lowest mark is: 56

Menu:
1. Find lowest mark
2. Find highest mark
3. Find average mark
4. Exit
Enter your choice: 2
The highest mark is: 90

Menu:
1. Find lowest mark
2. Find highest mark
3. Find average mark
4. Exit
Enter your choice: 3
The average mark is: 71.00

Menu:
1. Find lowest mark
2. Find highest mark
3. Find average mark
4. Exit
Enter your choice: 4
Exiting the program. Goodbye!
```


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Documentation

i) Copy and paste the C program from a compiler into Microsoft Word.

```
#include <stdio.h>

int s;
int student[100];
int arr;
int opt;
int i;
int stu;
void inputmarks();
void displaymark();
void highestmark();
void lowestmark();
void average();

int main()
{
   printf("Practical Test : DCBS 1 ( Hariz & Aiman )\n\n*********************************");
   inputmarks();
   printf("\n");
   menu();
}

void inputmarks()
{
   printf("\nEnter Number of students: ");
   scanf("%d", &s);
   printf("Enter the marks for each student:\n");

   for(int i = 0;i < s; i++)
   {
      printf("Enter marks for student %d: ", i + 1);
      scanf("%d", &student[i]);
   }

}

void menu()
{
     do {
   printf("\nMenu\n1. Find lowest mark\n2. Find highest mark\n3. Find average
mark\n4. Exit\n");
   printf("Enter your choice: ");
   scanf("%d", &opt);
```


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```
      switch(opt){

         case 1:
            lowestmark();
            break;
         case 2:
            highestmark();
            break;
         case 3:
            average();
            break;


     }

   }while ( opt != 4);
         printf("Exiting the program.. Thank you !");
         printf("\n");

}

void highestmark()
{
   int max = student[0];
   for(i = 0; i < s; i++)
   {
        if (student[i] > max)
         {
         max = student[i];
         }
   }
   printf("The highest mark is: %d\n", max);

}

void lowestmark()
{
   int min = student[0];
   for(i = 0; i < s; i++)
   {
        if (min > student[i])
         {
         min = student[i];
         }
   }
   printf("The lowest mark is: %d\n", min);
}
```


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```
void average()
{
   float total = 0;
   float avg = 0;
   for(i = 0;i < s; i++)
   {
      total = total + student[i];
      avg = total / s;
   }
   printf("The average mark is: %.2f\n", avg);

}
```

ii) Take a screenshot of the displayed output.

```
Practical Test : DCBS 1 ( Hariz & Aiman )

******************************
Enter Number of students: 2
Enter the marks for each student:
Enter marks for student 1: 51
Enter marks for student 2: 20


Menu
1. Find lowest mark
2. Find highest mark
3. Find average mark
4. Exit
Enter your choice: 1
The lowest mark is: 20

Menu
1. Find lowest mark
2. Find highest mark
3. Find average mark
4. Exit
Enter your choice: 2
The highest mark is: 51

Menu
1. Find lowest mark
2. Find highest mark
3. Find average mark
4. Exit
Enter your choice: 3
The average mark is: 35.50

Menu
1. Find lowest mark
2. Find highest mark
3. Find average mark
4. Exit
Enter your choice: 4
Exiting the program.. Thank you !
```

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iii) Upload the C program file.

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